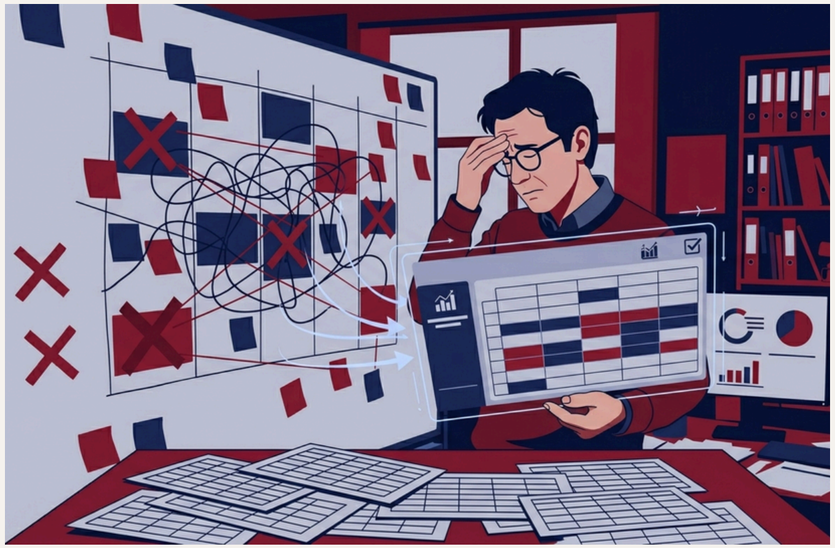
Title: Intelligent Timetable Generator for Multi-Section Academic Scheduling

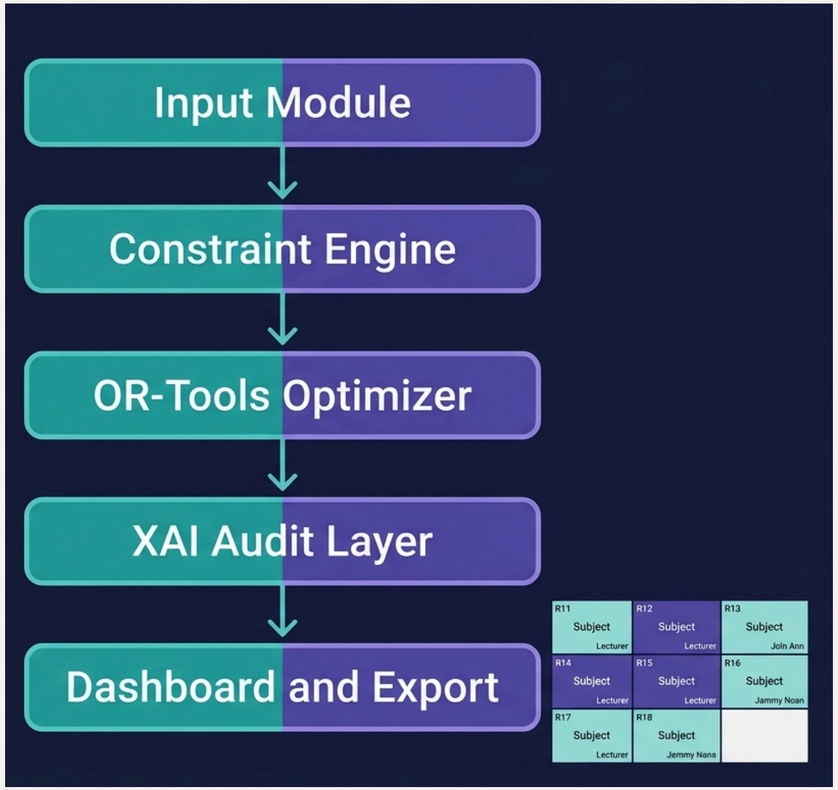
Team: Binary Brains | GD Goenka University | Problem Statement J | Smart Campus Hackathon 2026

PROBLEM UNDERSTANDING

- Manual timetabling takes 3-7 days per semester using spreadsheets
- Faculty double-booking, room overlap, and wrong lab assignments are common
- Lab sessions need consecutive periods - impossible to manage manually without errors
- Faculty teach multiple subjects across different rooms and sections simultaneously
- Zero visibility no room number, lecturer name, or class info inside timetable slots



Proposed Solution Architecture



Input : Subject codes, faculty list, room types, sections, combined classes, number of periods per day, lunch block

Constraint Engine : enforces no double-booking, lab-only practicals, room capacity, lunch break

OR-Tools Optimizer: Generates 100% conflict-free timetable automatically

XAI Audit Layer : logs why every slot was assigned

Auto Faculty Timetable : personal weekly schedule generated for every lecturer instantly

Cell Design : each slot shows Room No. bottom-left, Subject Code center, Lecturer Name bottom-right

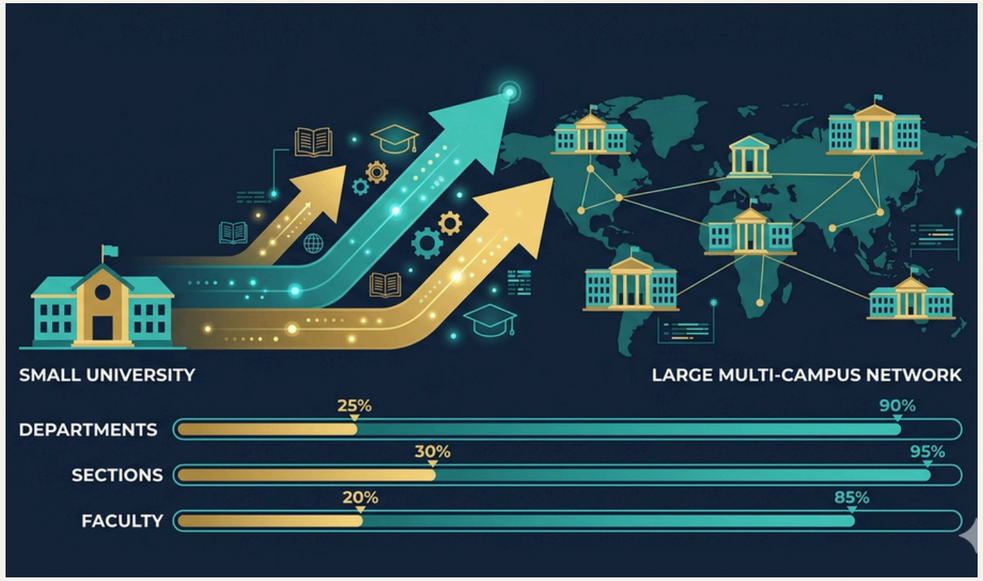
Output : section-wise, faculty-wise, room-wise views + PDF/Excel export

Technology Stack

- **Frontend :** Next.js, Tailwind CSS, Chart.js.
- **Backend :** Python FastAPI, Google OR-Tools, Pandas.
- **Database :** Supabase (PostgreSQL).
- **Deployment :** Vercel (Frontend), Render (Backend).
- **Version Control :** GitHub.

- Supabase stores subject codes, room numbers, faculty names, period config and constraints
- OR-Tools handles lab consecutive-period grouping, lunch break blocking, and faculty conflict resolution

Scalability Strategy



- Modular design - new departments and courses added without changing core logic
- Supports all room types theory rooms and lab rooms assigned automatically by subject type
- Supabase handles multi-semester and multi-campus data
- Stateless FastAPI scales to 50+ sections and 200+ faculty simultaneously
- Role-based access students see section view, faculty see only their own schedule, admin sees full master timetable
- Extendable to attendance, leave management, and academic planning systems

Expected Institutional Impact

Metric Value :

Scheduling time reduced	**95%**
Conflicts after generation	**0**
Faster disruption recovery	**3×**
Slot decisions auditable	**100%**

	1 8:00 - 8:50	2 9:00 - 9:50	3 10:00 - 10:50	4 11:00 - 11:50	5 12:00 - 12:50	6 13:00 - 13:50	7 14:00 - 14:50	8 15:00 - 15:50	9 16:00 - 16:50
Mo	**CS101** B201 JSM			**MT103** C103 SAM		**CS101L** L102 JSM			**PY102** A305 KLT
Tu	**HS204** D101 DNV		**PY102** A305 KLT		**MT103** C103 SAM	**CH205** B201 JBR		**EL206** C103 RPK	
We	**CS101** B201 JSM	**MT103** C103 SAM		**PY201L** L104 KLT			**CH205** B201 JBR		
Th	**EL206** C103 RPK		**EE301L** L106 GKP			**HS204** D101 DNV			
Fr	**PY102** A305 KLT	**CS101** B201 JSM		**MT103** C103 SAM	**HS204** D101 DNV		**MT103 TUT** C103 SAM		

Timetable generated: 2024-10-15

- Timetable generated in under 60 seconds replaces a 3–7 day manual process
- Every cell shows subject code, room number, and lecturer name zero ambiguity for students
- Lab sessions auto-scheduled as consecutive multi-period blocks no manual grouping needed
- Faculty personal timetable auto-generated each lecturer sees only their own weekly schedule
- One generation produces views for students, faculty, and admin simultaneously